

PROPOSTA DI GRANT GIOVANI IN CSN5 - Anno 2026

Acronimo:	PRISM
Titolo completo:	Precision Room-temperature Instrumentation for high-energy x-ray absorption Spectroscopy Measurements
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Abstract (2000 caratteri)

X-ray absorption spectroscopy (XAS) is a powerful element-selective technique for probing the electronic and atomic structure of materials. While synchrotron and free-electron laser facilities provide outstanding performance, their limited accessibility motivates advanced laboratory instrumentation. A major challenge is extending fluorescence-mode XAS beyond 20 keV, where silicon detectors lose efficiency and germanium detectors require cryogenic operation. This limits studies of elements such as Mo, Rh, Ag and Cd, whose K-edges are relevant to catalysis, electrochemistry, advanced alloys and energy materials. PRISM will develop a high-resolution laboratory hard X-ray spectrometer combining advanced crystal optics with a multichannel cadmium zinc telluride (CZT) detector for precision fluorescence spectroscopy up to 50 keV. Building on my expertise in crystal-based X-ray spectroscopy gained within the VOXES project and SIDDHARTA-2 experiment, PRISM will develop detector architectures, low-noise front-end electronics, calibration procedures and spectral reconstruction algorithms to achieve sub-keV energy resolution, high quantum efficiency (>90% up to ~40 keV) and excellent SNR performance at room temperature. The instrument will be validated through fluorescence XAS measurements on reference materials with K-edges above 20 keV, demonstrating accurate edge reconstruction and XANES spectra within laboratory-compatible acquisition times. The resulting capability will enable extended in situ and operando studies of slowly evolving catalytic, electrochemical and energy-related processes. Beyond this proof of concept, PRISM will establish a new detector platform for next-generation laboratory hard X-ray spectroscopy, strengthening INFN expertise in semiconductor radiation detectors, precision X-ray instrumentation and crystal spectroscopy. The technology will complement future photon-source infrastructures such as EuAPS and support research and industrial applications.

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1 State of the art (4000 caratteri)

A central challenge in materials characterization is determining how electronic and local atomic structure evolve under realistic conditions. Changes induced by irradiation, processing, phase transformations, chemical reactions or device operation may be transient, reversible or environment-dependent, and can therefore be missed by ex-situ measurements [1]. This requires in-situ and/or operando techniques capable of probing thick, heterogeneous or diluted samples while retaining sensitivity to selected elements and their local coordination [2,3]. Hence, there is a broad need for non-destructive, element-selective and bulk-sensitive probes compatible with complex environments and long acquisition campaigns.

X-rays are particularly suited for such investigations because of their high penetration depth with respect to other optical techniques, enabling access to the bulk of samples. By tuning the photon energy of a selected element, X-ray spectroscopies provide element- and orbital-specific information, even in chemically complex and heterogeneous systems. These properties make X-ray methods well-suited to non-destructive in situ and operando studies [4].

Among X-ray techniques, X-ray absorption spectroscopy (XAS) is particularly suited to this purpose because it measures the energy-dependent absorption coefficient across an element-specific absorption edge [5,6]. The near-edge region, the XANES, is sensitive to oxidation state, electronic structure and coordination geometry, whereas the extended region up to 1 keV above the edge, the EXAFS, provides quantitative information on the local atomic environment, including interatomic distances, coordination numbers and structural disorder. XAS can be measured in transmission, by recording the attenuation of the incident beam through the sample, or in fluorescence, by detecting the characteristic X-rays emitted after absorption. While transmission is generally preferred for homogeneous samples of suitable thickness, fluorescence detection is essential for diluted absorbers, thick specimens and complex sample environments, where the transmitted intensity is too weak or difficult to access.

Synchrotron facilities are the reference platforms for XAS, as their high brilliance, broad energy tunability and dedicated beamlines enable high-quality spectra with excellent signal-to-noise ratio over a wide energy range. X-ray free-electron lasers further provide ultrashort and intense X-ray pulses, opening access to femtosecond time-resolved experiments and transient states. However, access to these large-scale facilities is limited and competitive, and the allocated beamtime is often unsuitable for routine characterization, long experimental campaigns, or studies involving hazardous and radioactive samples [7].

These limitations have driven renewed interest in laboratory XAS systems based on conventional X-ray tubes [8]. Although laboratory sources provide substantially lower brilliance than synchrotrons, they offer continuous access, flexible sample handling and long measurement times, making them attractive for routine analysis, method development and slowly evolving in situ or operando processes. Recent advances in source brightness, crystal optics and detection technology have enabled an increasing number of XAS and XES measurements to be transferred back to the laboratory.

Within INFN, VOXES [9] has established expertise in high-resolution von Hamos spectroscopy for extended and dilute X-ray sources. The instrument currently operates mainly in the 5–20 keV range and has recently been upgraded with motorized positioning, energy-dispersive monitoring and mechanical compatibility with laboratory XAS measurements [10].

Despite this progress, laboratory fluorescence-XAS at energies above approximately 20 keV remains challenging. Silicon-based detectors progressively lose effectiveness in the hard-X-ray range, whereas high-purity germanium detectors offer excellent spectroscopic performance but require cryogenic cooling. Conversely, high-Z compound semiconductors such as CdTe and CdZnTe provide high

hard-X-ray absorption efficiency and room-temperature operation, although their use requires careful control of charge transport, detector segmentation, electronic noise and spectral reconstruction.

As a consequence, the K-edges of technologically relevant elements such as Mo, Rh, Ag and Cd remain difficult to investigate by laboratory fluorescence-XAS under realistic experimental conditions. There is therefore a clear need for a compact platform combining high-resolution crystal optics with efficient, room-temperature hard-X-ray detection. Addressing this technological gap would extend laboratory XAS beyond 20 keV and establish a complementary capability between conventional laboratory instruments, synchrotron facilities and future compact photon sources.

2 Goals and Expected Outcome (2000 caratteri)

The **main goal** of the project is **to develop and validate a room-temperature laboratory platform for fluorescence-mode X-ray absorption spectroscopy (f-XAS) above 20 keV**. Source, crystal optics, sample geometry, CZT detection, electronics and simulations will be optimized as a single measurement chain to obtain chemically meaningful XANES spectra from samples for which transmission is impractical.

The project is organized into the following specific **goals (G)** and **expected outcomes (EO)**:

G1 – Establish a validated design and predictive model of the complete instrument.

EO1 – An integrated simulation workflow and a frozen experimental design, validated at the Ag K-edge and at one energy of at least 40 keV. Predictions and measurements will agree within 20% for detected rate and within 10% for spot size and energy resolution.

G2 – Develop and characterize a multichannel CZT detector operating at room temperature.

EO2 – A calibrated system with intrinsic efficiency above 90% at 40 keV, energy resolution of 1 keV FWHM or better over 20–40 keV, dead time below 10% and residual pile-up below 5% at the selected operating rate.

G3 – Validate the complete platform on reference compounds with different chemical states.

EO3 – Quantitative Ag K-edge XANES with effective XAS resolution of 10 eV FWHM or better, edge-position reproducibility within 1 eV, edge-step SNR of at least 20 and acquisition time below the 12 h laboratory benchmark. The normalized spectral deviation from reference data will be below 5%.

G4 – Demonstrate f-XAS on at least one thick, diluted or operando-compatible sample.

EO4 – Discrimination of at least two chemical states through an edge shift of at least 1 eV and/or a normalized white-line variation of at least 5%, with a significance of at least 3σ and an acquisition time below 12 h per spectrum. The measured operating domain will be reported in terms of concentration, sample geometry and acquisition time.

3 Methodology (10000 caratteri)

4 Organisation (4000 caratteri)

dd

5 Synergies with Other Activities (3000 caratteri)

EuAPS

VOXES

PANDORA

6 Impact and Possible Spillovers (3000 caratteri)

d

7 Costs

-X-ray microfocus tube

-Crystals

-Axis

-Sample targets

-EXRS2028 Paris?

Year			
CZT			

8 Risk Assessment

9 References

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